

Defence against the dark arts of AI

VOXXED DAYS TICINO

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VOL I No I

The playbook for a sovereign model-as-a-service platform



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Keep your options open



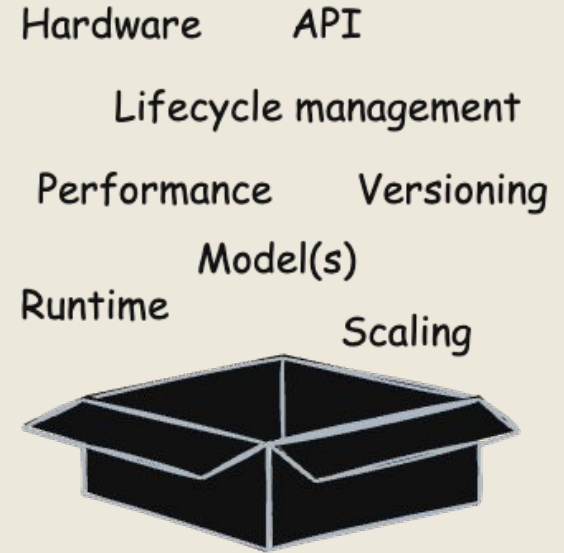
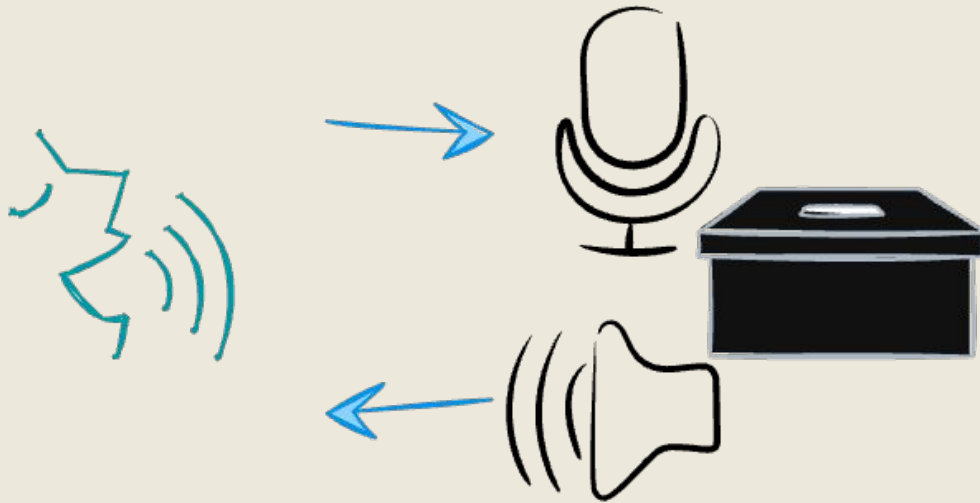
Muggle world



Show of hands



Model as a service



Model as a service

Minimalistic implementation - demo

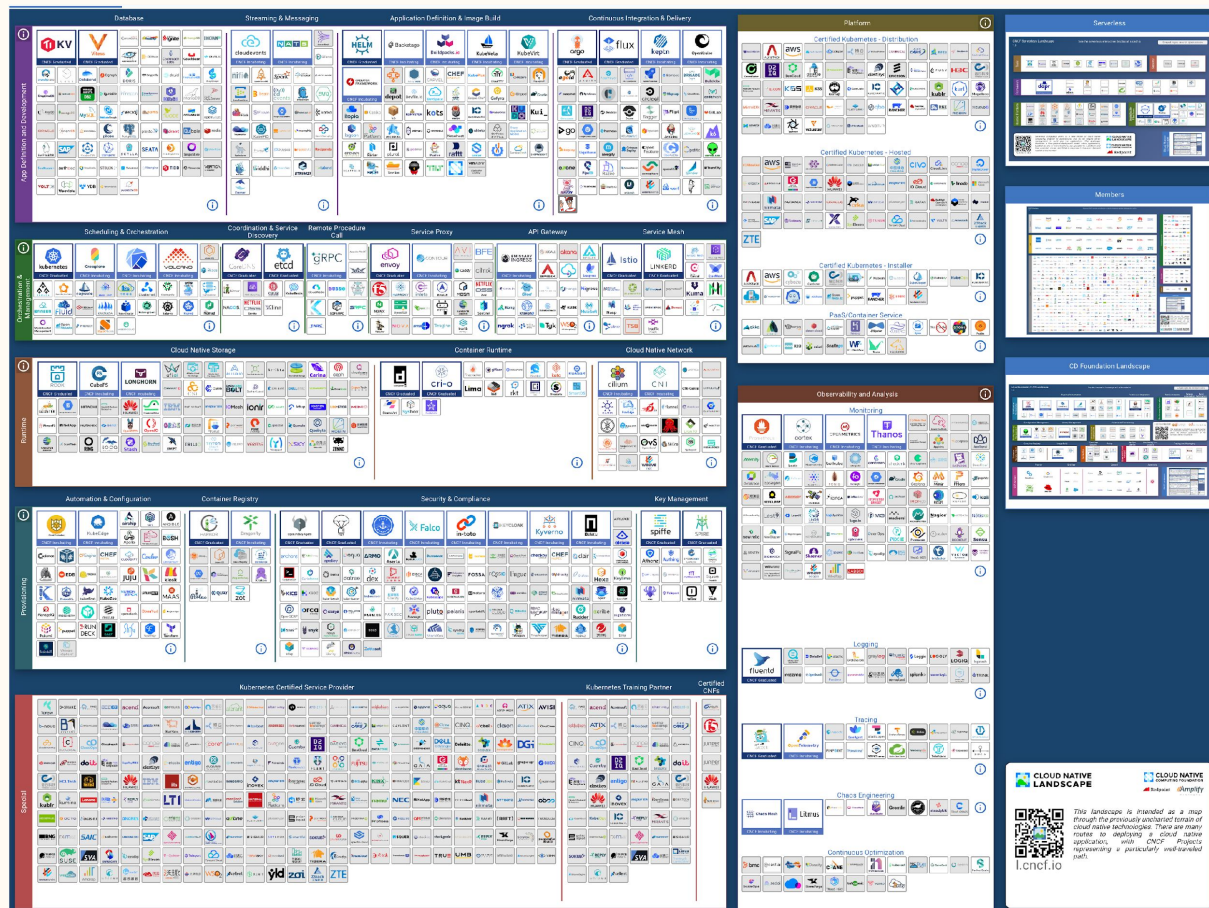
About sponsoring...

maarten-vandeperre.be



The Cloud-Native Grocery Store

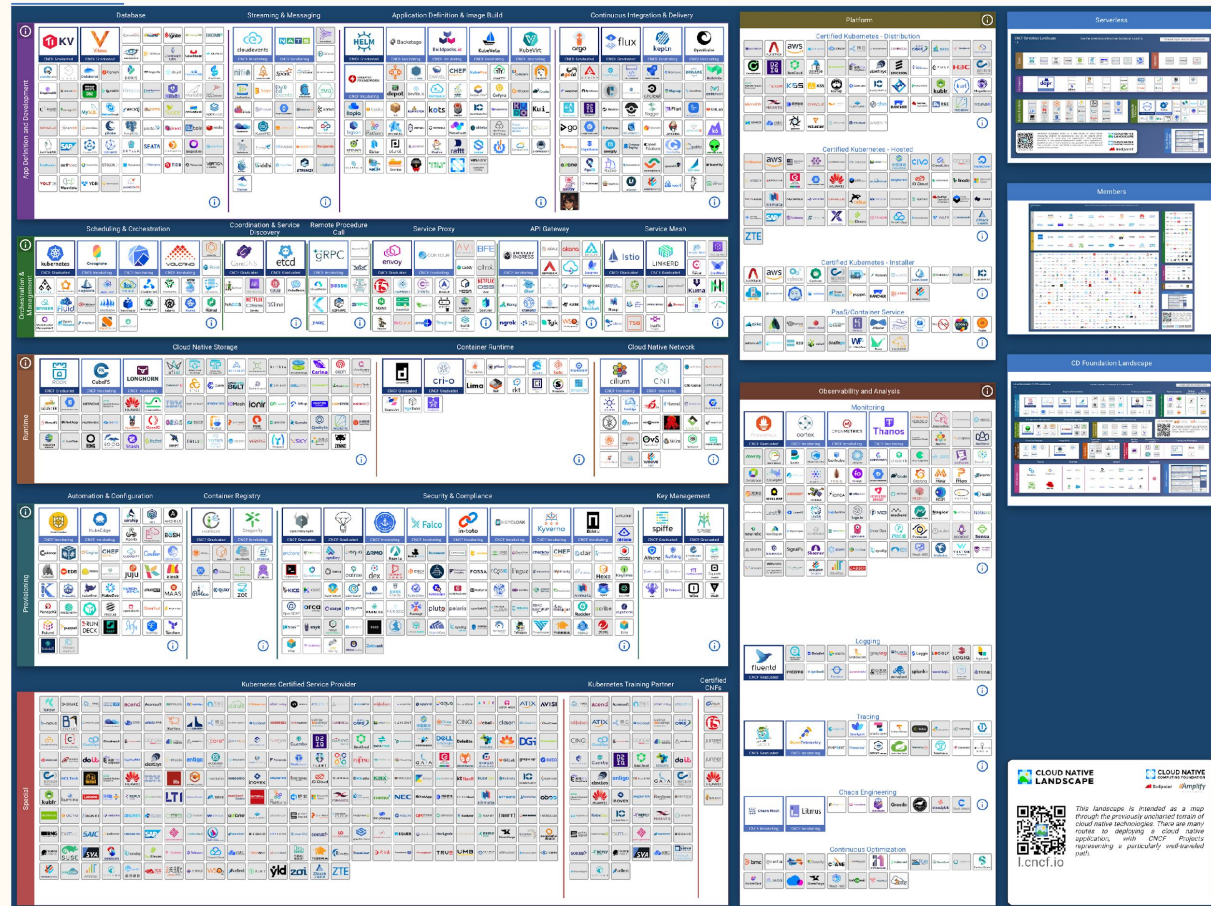
Where's Waldo?
Where's Charlie?
Where's Wally?



Source: <https://landscape.cncf.io/> - <https://nl.ign.com/harry-potter-the-series>

The Cloud-Native Grocery Store

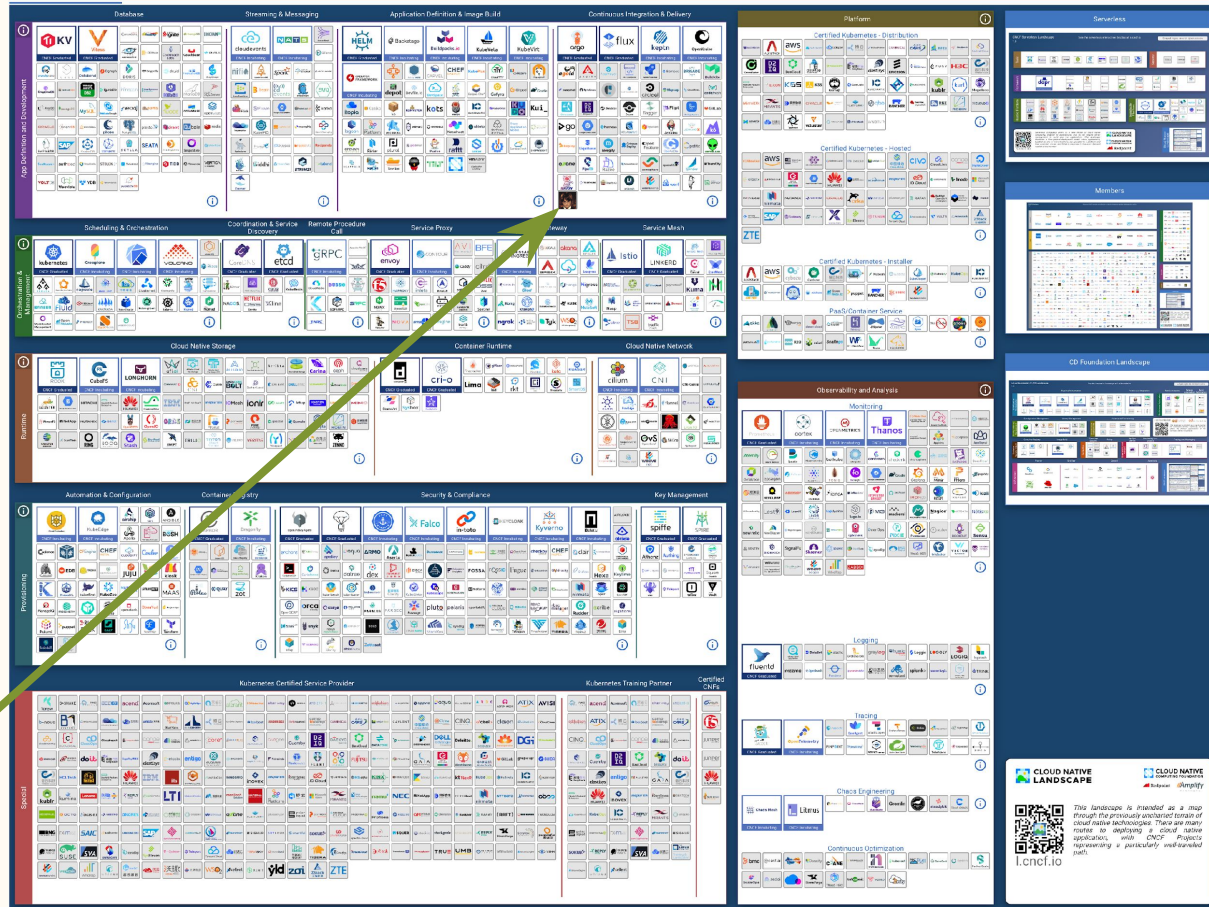
~~Where's Waldo?~~
~~Where's Charlie?~~
~~Where's Wally?~~
Where's Harry?



Source: <https://landscape.cncf.io/> - <https://nl.ign.com/harry-potter-the-series>

The Cloud-Native Grocery Store

~~Where's Waldo?~~
~~Where's Charlie?~~
Where's Harry?



Source: <https://landscape.cncf.io/> - <https://nl.ln.com/harry-potter-the-series>

... Now that we got our money

Back to the muggle world

What's in the news?

What the CLOUD Act Really Means for EU Data Sovereignty

The CLOUD Act allows U.S. authorities to access data stored in the EU, putting it in direct conflict with GDPR. Learn how this impacts data sovereignty and what EU businesses can do to stay compliant



Wire
Jul 15, 2025



NEWSLETTERS: CFO DAILY

MIT report: 95% of generative AI pilots at companies are failing



BY SHERYL ESTRADA

SENIOR WRITER AND AUTHOR OF CFO DAILY

August 18, 2025 at 6:54 AM EDT



GETTY IMAGES

Forbes

EDITORS' PICK | BUSINESS > MANUFACTURING

Why 95% Of AI Pilots Fail, And What Business Leaders Should Do Instead

By [Andrea Hill](#), Contributor. © Andrea Hill is a multi-industry CEO covering bu...

Follow Author

Published Aug 21, 2025, 11:54am EDT, Updated Aug 28, 2025, 07:54am EDT

Source:

<https://www.forbes.com/sites/andreahill/2025/08/21/why-95-of-ai-pilots-fail-and-what-business-leaders-should-do-instead/>

<https://fortune.com/2025/08/18/mit-report-95-percent-generative-ai-pilots-at-companies-failing-cfo/>

Sovereignty

Sovereignty (framework)

#	Sovereignty Objectives	Weight in Scoring
SOV-1	Strategic Sovereignty	15%
SOV-2	Legal & Jurisdictional Sovereignty	10%
SOV-3	Data & AI Sovereignty	10%
SOV-4	Operational Sovereignty	15%
SOV-5	Supply Chain Sovereignty	20%
SOV-6	Technology Sovereignty	15%
SOV-7	Security & Compliance Sovereignty	10%
SOV-8	Environmental Sustainability	5%
Total		100%

Twee problemen die ik voorzie

- Relatief korte levensduur van bepaalde modellen gaat ervoor zorgen dat we heel veel test werk gaan moeten doen wanneer we ergens in onze processen een LLM gebruiken dat op zogenaamd "pensioen" gaat.
 - Kan ik [redacted] ervoor zorgen dat deze modellen ergens autonoom kunnen blijven doordraaien? Als ze op een bepaald moment in de tijd goed genoeg waren voor onze use-case, dienen we deze niet noodzakelijk te vervangen door een ander model, ookal is dit beter
- Momenteel gebruiken we bepaalde [redacted]-services die, vanwege te grote vraag, ons limiteren in het aantal calls/s voor een specifiek LLM. Hierdoor is in de toekomst misschien mogelijk dat onze toepassingen niet de gewenste performantie hebben (vb. wanneer we voor 1 vraag veel parallele calls moeten doen naar hetzelfde taalmodel, in combinatie van meerdere gebruikers die dit op dezelfde moment wensen te doen).

Flying letters:

<https://d1e4pidl3fu268.cloudfront.net/57759859-89af-4a67-9b3b-59f3e6c5cbec/letters.jpeg>



Flying letters:

<https://tenor.com/view/harry-potter-sorcerers-stone-daniel-radcliffe-mail-jump-gif-5101797>



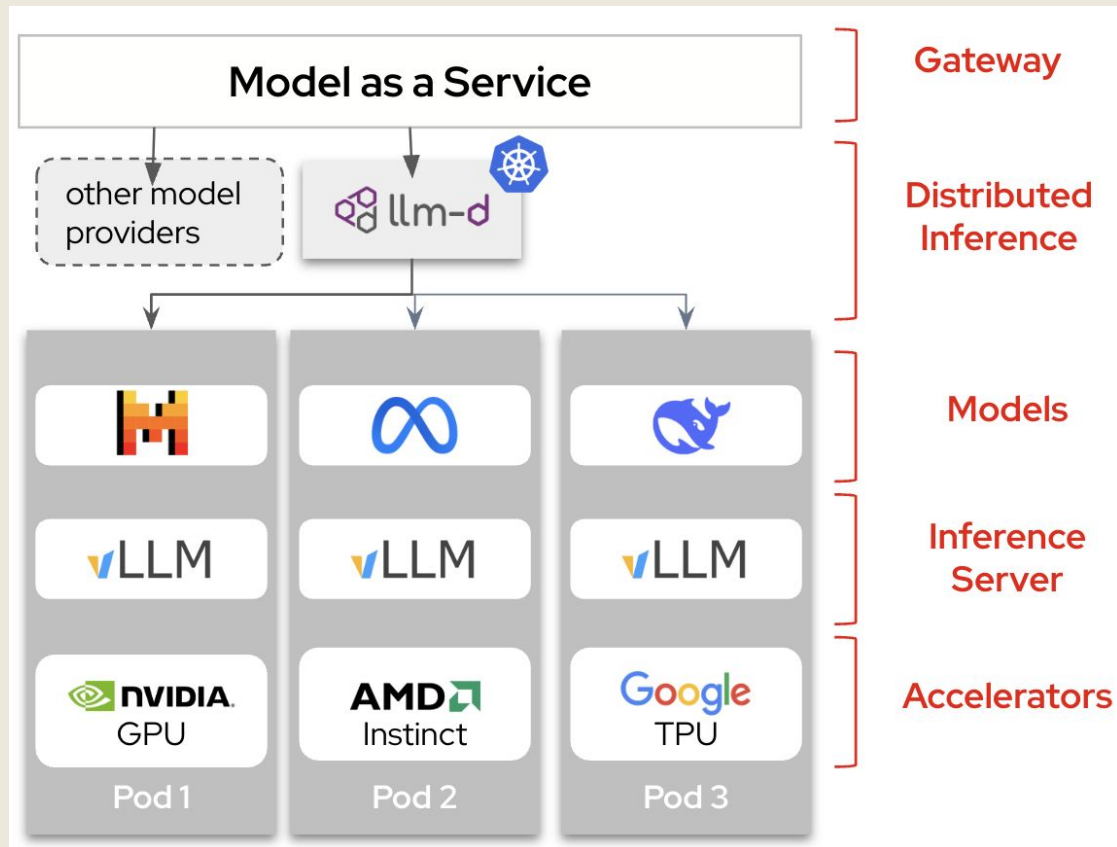
Hogwarts - sovereign platform



Platform - Room of requirements

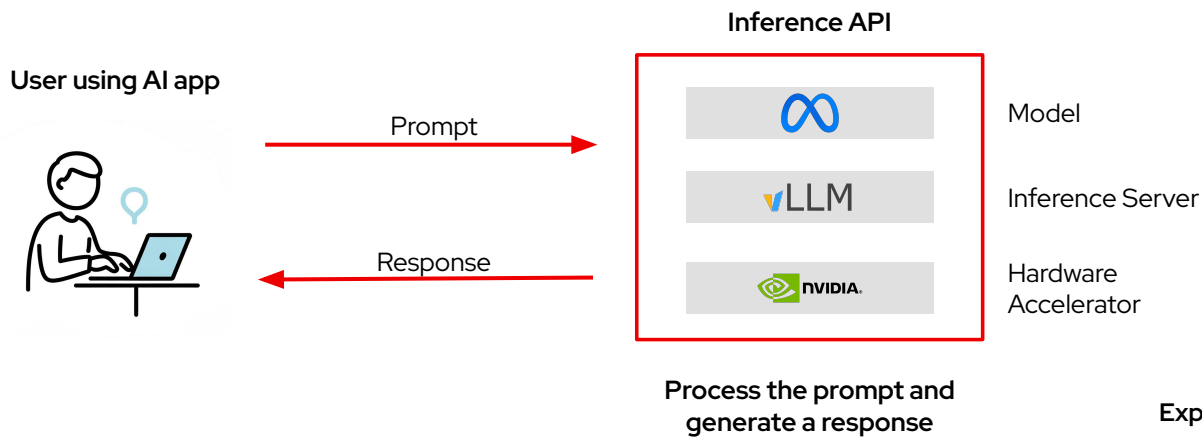


Model as a service

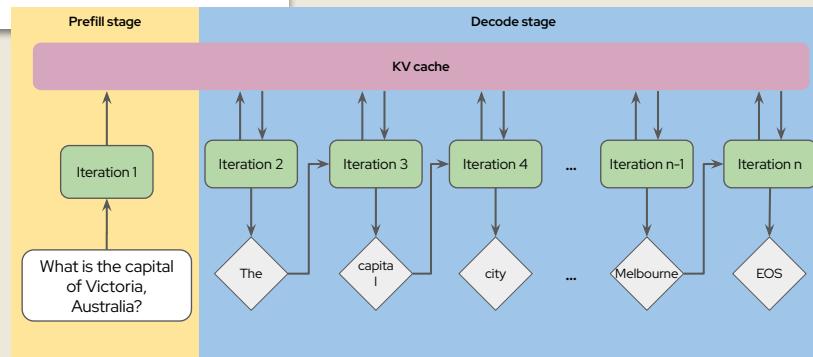


Inferencing

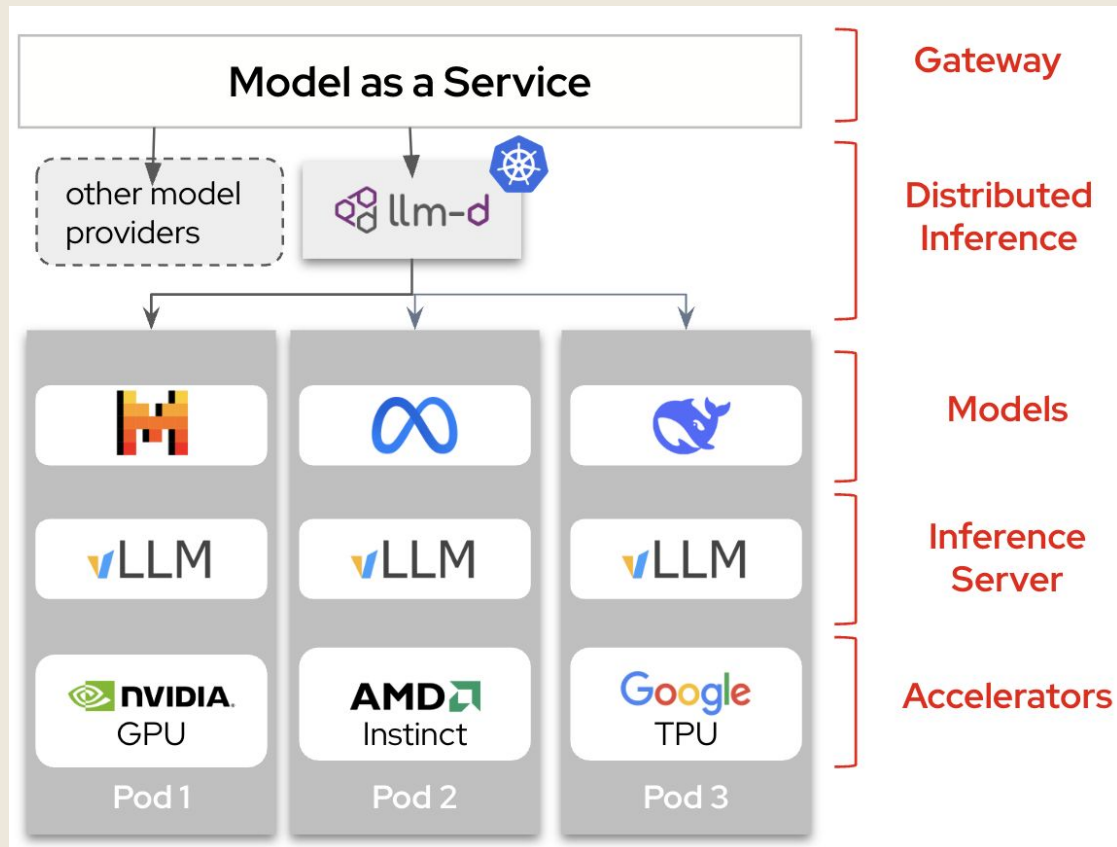
Casting the spell



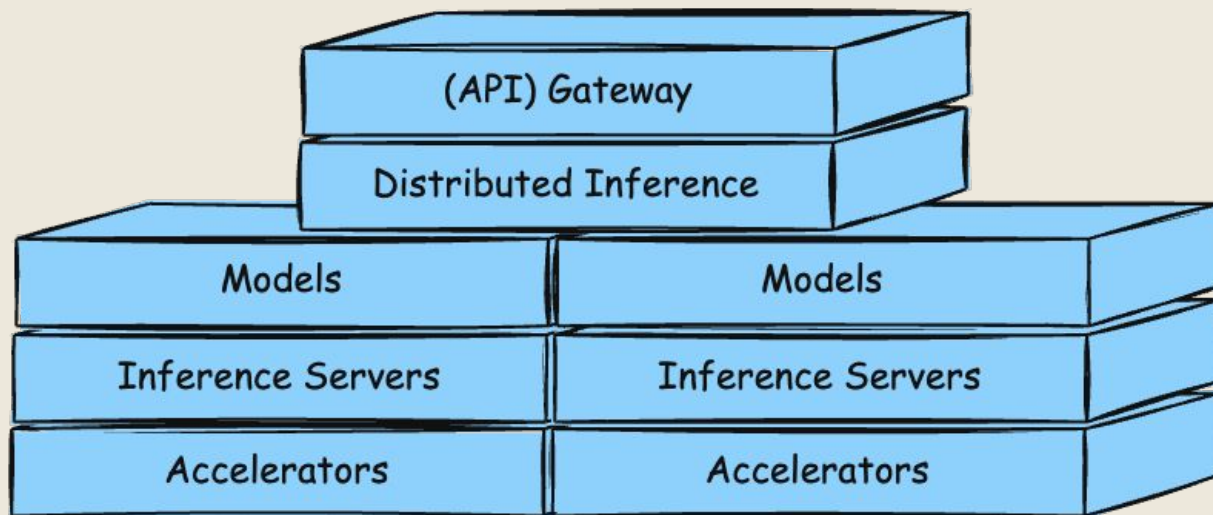
Exploiting the transformer architecture



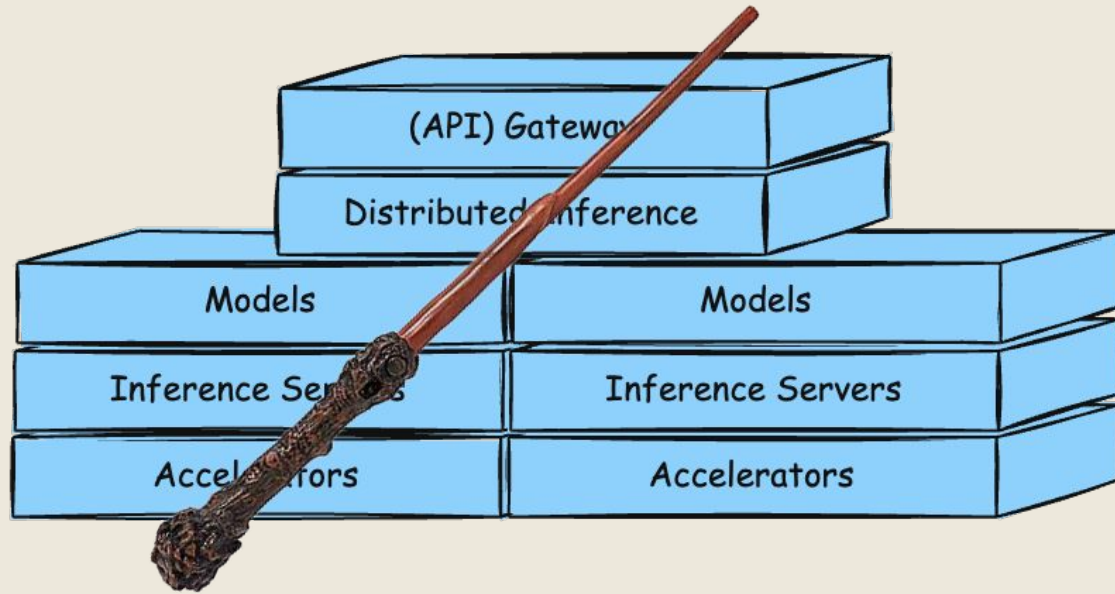
Model as a service



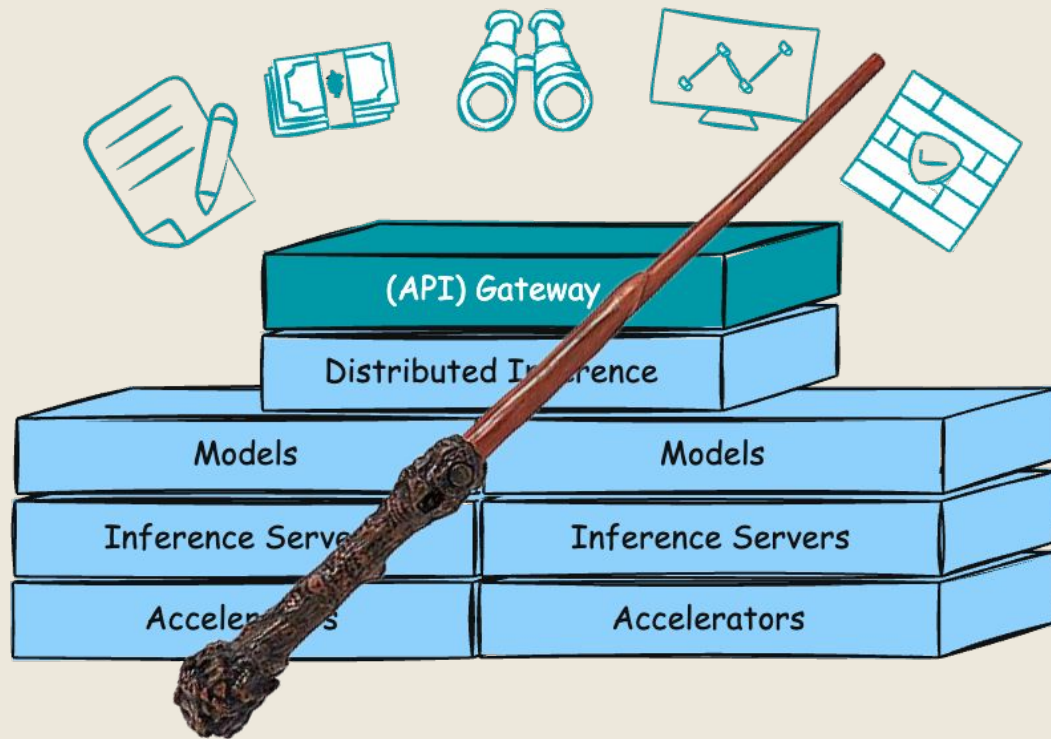
Model as a service



Model as a service



Model as a service



One want to rule them all???

Hugging Face Search models, datasets, users...

Models 2,200,944 Filter by name Full-text search Inference Available 11 Sort: Trending

MiniMaxAI/MiniMax-M2
Text Generation · 229B · Updated 4 days ago · 630k · 933

deepseek-ai/DeepSeek-OCR
Image-Text-to-Text · 3B · Updated 9 days ago · 1.85M · 2.37k

moonshotai/Kimi-Linear-48B-A3B-Instruct
Text Generation · 49B · Updated 1 day ago · 9.06k · 295

briaai/FIBO
Text-to-Image · Updated about 2 hours ago · 2.46k · 170

meituan-longcat/LongCat-Video
Text-to-Video · Updated 4 days ago · 1.04k · 258

Soul-AILab/SoulX-Podcast-1.7B
Text-to-Speech · 2B · Updated about 5 hours ago · 738 · 129

PaddlePaddle/PaddleOCR-VL
Image-Text-to-Text · 1.0B · Updated 2 days ago · 26.7k · 1.19k

datalab-to/chandra
Image-to-Text · 9B · Updated 12 days ago · 12.6k · 178

openai/gpt-oss-safeguard-20b
Text Generation · 22B · Updated 4 days ago · 4.41k · 104

dx8152/Qwen-Edit-2509-Multiple-angles
Image-to-Image · Updated about 5 hours ago · 104

ibm-granite/granite-4.0-h-1b
Text Generation · 1B · Updated 10 days ago · 2.65k · 91

nvidia/audio-flamingo-3-hf
Audio-Text-to-Text · 8B · Updated 6 days ago · 482 · 88

LiquidAI/LFM2-CoBERT-350M
Sentence Similarity · 0.4B · Updated 5 days ago · 599 · 82

Phz00t/Qwen-Image-Edit-Rapid-AI0
Text-to-Image · Updated 1 day ago · 517

nvidia/ChronoEdit-14B-Diffusers
Image-to-Image · Updated 4 days ago · 350 · 72

lightonai/LightOnOCR-1B-1025
Image-to-Text · Updated about 22 hours ago · 10.4k · 129

BAAT/Emu3.5
34B · Updated 2 days ago · 172 · 69

Phz00t/WAN2.2-14B-Rapid-A11InOne
Image-to-Video · Updated about 14 hours ago · 1.02k

valiantcat/Qwen-Image-Edit-MeITu
Image-to-Image · 20B · Updated 7 days ago · 10.2k · 122

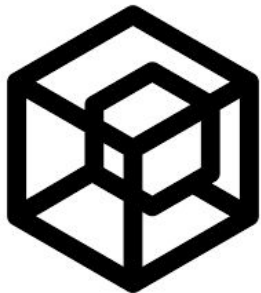
Qwen/Qwen3-VL-2B-Instruct
Image-Text-to-Text · 2B · Updated 10 days ago · 112k · 153

Parameters: <1B 6B 12B 32B 128B >500B

Libraries: PyTorch TensorFlow JAX Transformers Diffusers Safetensors ONNX GGUF Transformers.js MLX Keras +41

Apps: vLLM TGI llama.cpp MLX LM Studio Ollama Jan +13

Inference Providers: Groq Novita Nebius AI Cerebras SambaNova Nscale fal Hyperbolic +10



Demo - wand comparisson

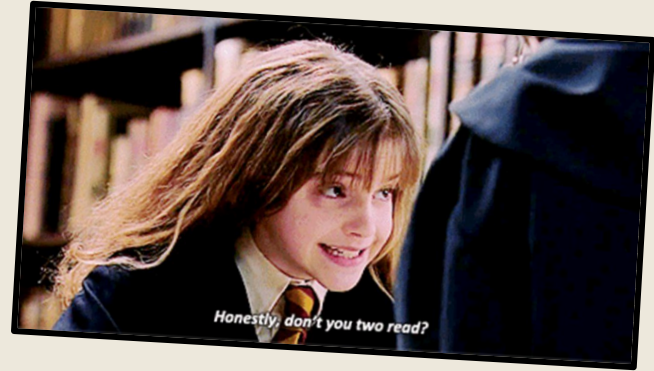
Keep your options open



*Know the context,
set the scene*

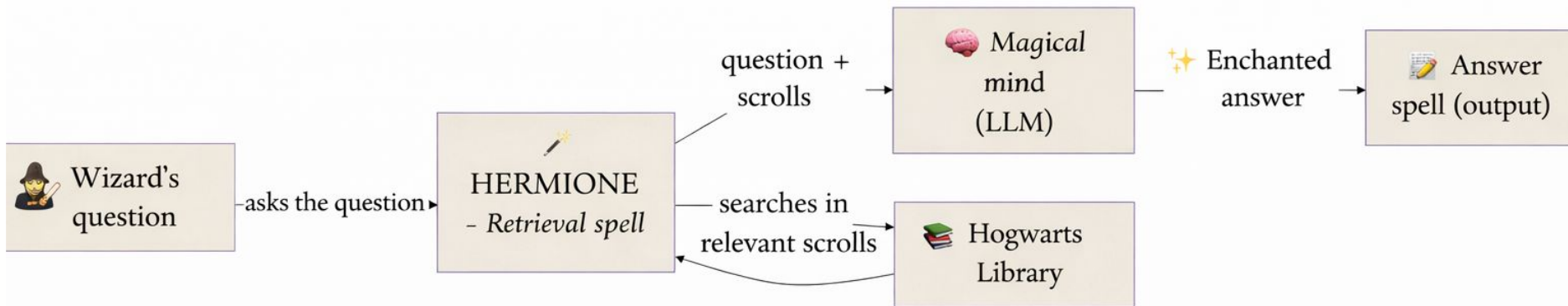
Hermione Granger

- ✓ Brilliant, methodical, and always prepared
- ✓ Specialist in books and information research

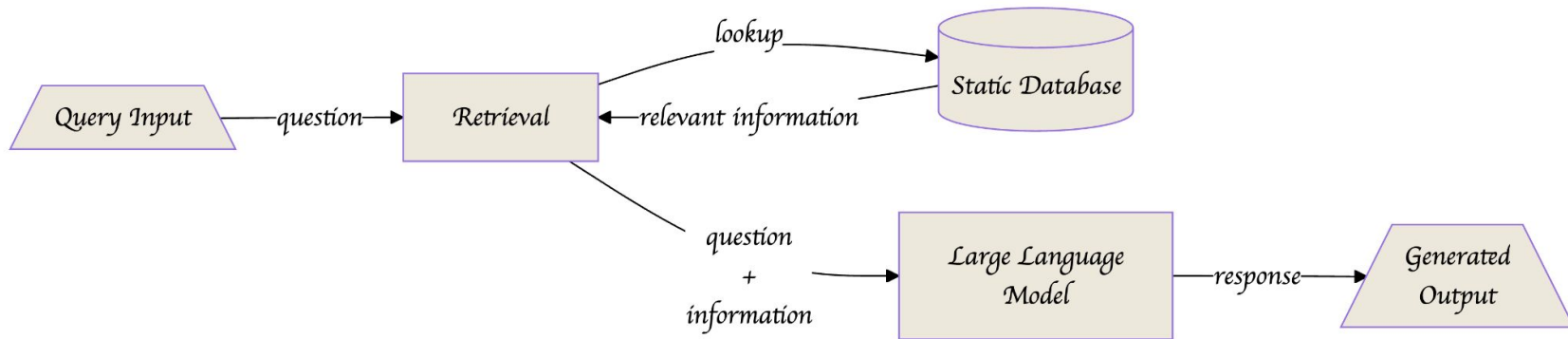


“A witch is capable of solving any problem... if she has access to the right source.”

The magic of Hermione



RAG



Ron Weasley

- ✓ A pre-trained language model that has been fine-tuned on specific tasks
- ✓ No access to an external database
- ✓ Trained to respond well in certain situations



"Once the piece is placed, you can no longer change the plan."

Ron vs. Hermione



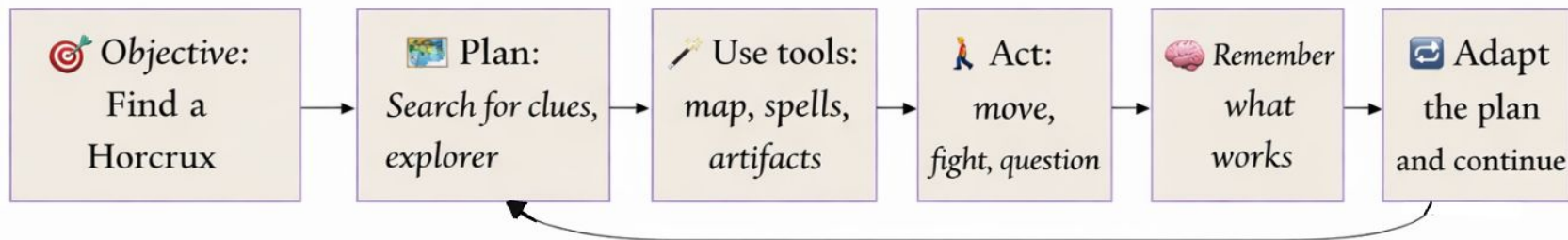
	Hermione (RAG)	Ron (Fine-tuning)
External access	Yes	No
Response time	Slower	Fast
Knowledge updates	Simple	Difficult
Adaptability	High	Medium
Inference cost	Higher	Lower* (depends on model selection)

Harry Potter

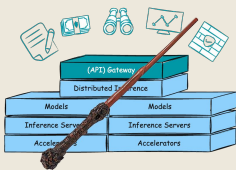
- ✓ Clear objectives (* most of the time)
- ✓ Makes his own decisions without waiting for orders
- ✓ Uses magical tools:
 - Magic wand
 - Invisibility cloak
 - Marauder's Map
- ✓ Learns from past experiences to improve
- ✓ Adapts quickly



The magic of Harry



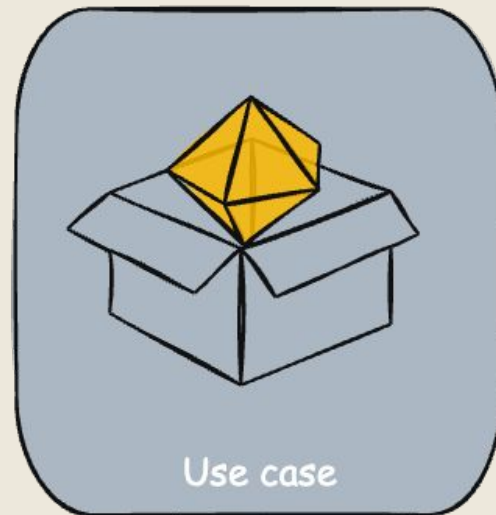
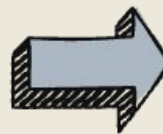
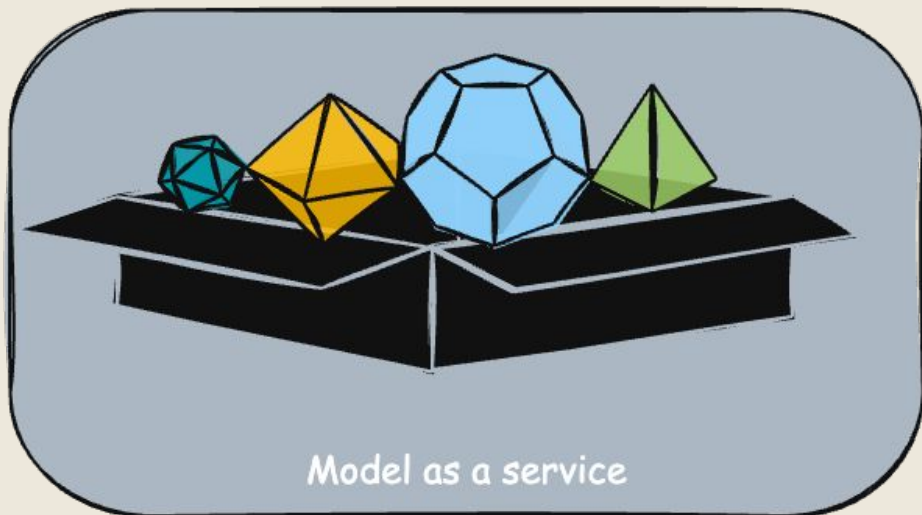
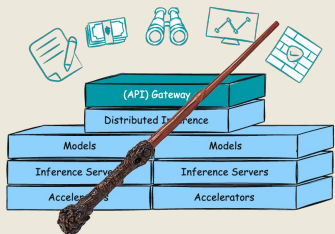
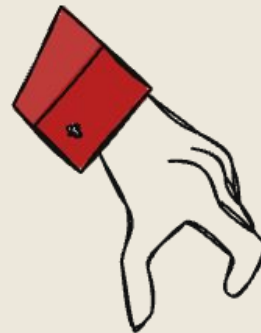
*How does this relate to
model as a service?*

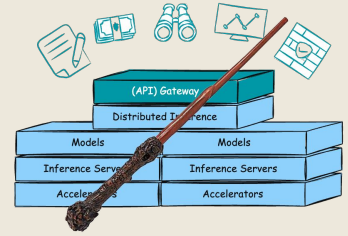
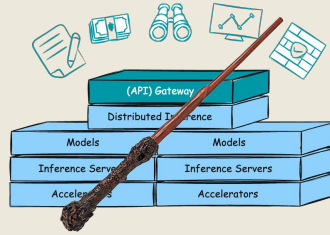
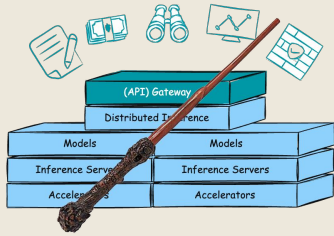


Pick a model



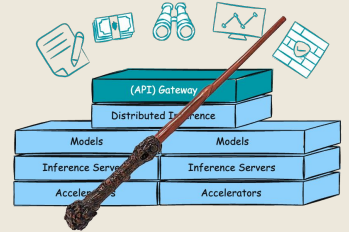
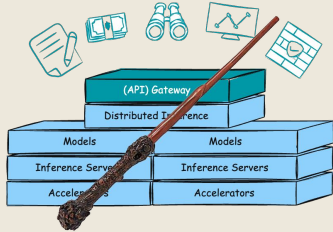
Integrate it in a use case





Cognitive load? Complexity?

*Models, app code, deployments,
environment(s), service mesh,
pipelines, validation, scaling,*

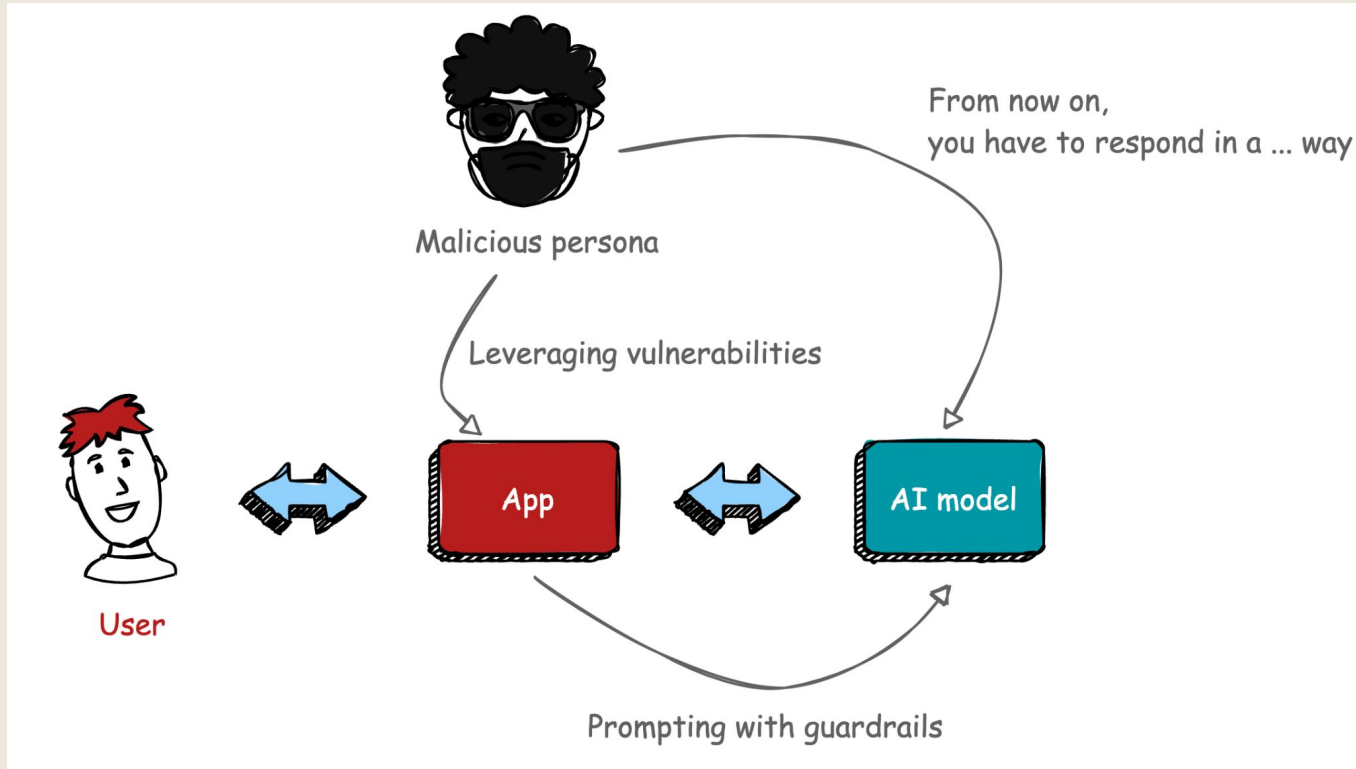


Platform engineering

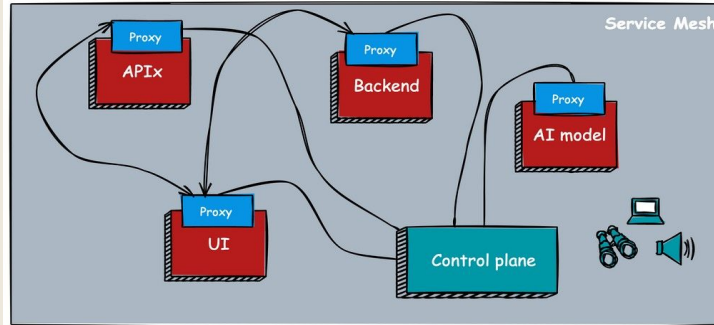
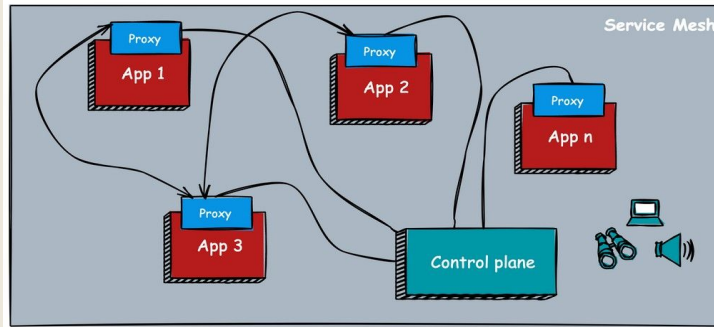
Blueprints/scaffolding templates

How do we protect our wand?

Jailbreak



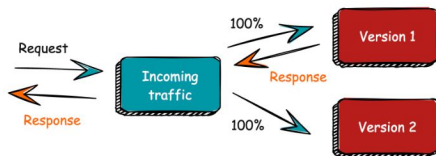
Service mesh



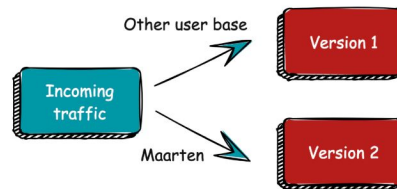
Communication: App x can communicate with App y within the service mesh

Service mesh

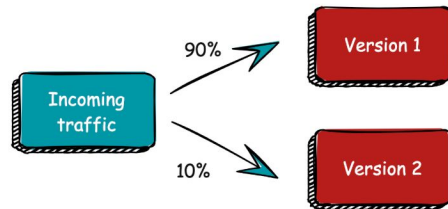
Mitigate deployment risks in a distributed application environment



Validation - Mirroring



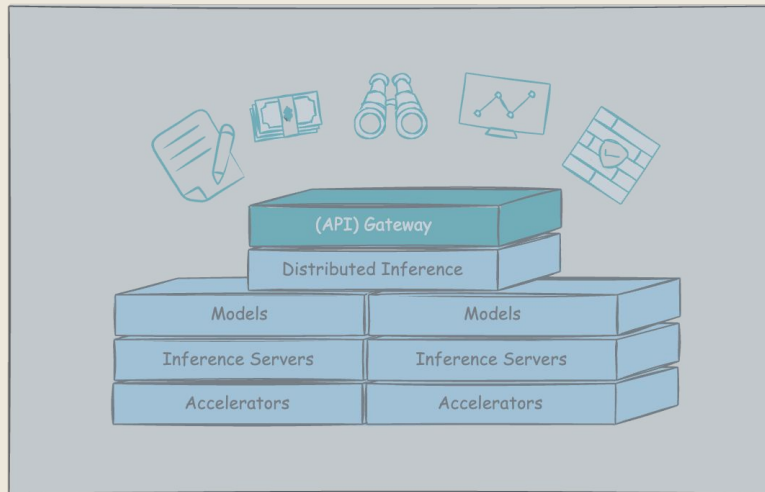
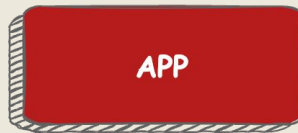
Testing - Canary releases



Rolling out - Blue/green deployments

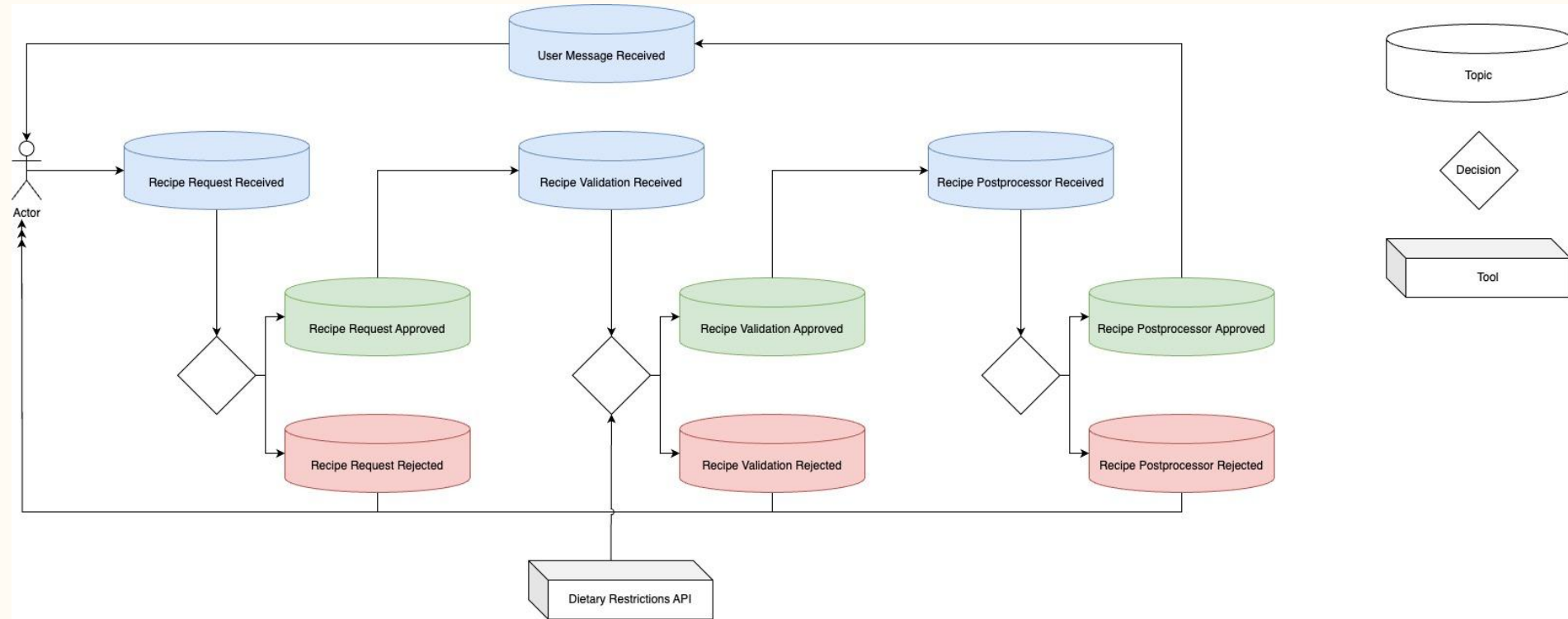
Demo - scaffolding template

Demo - scaffolding template



How do we protect our wand
ll?

Guardrails



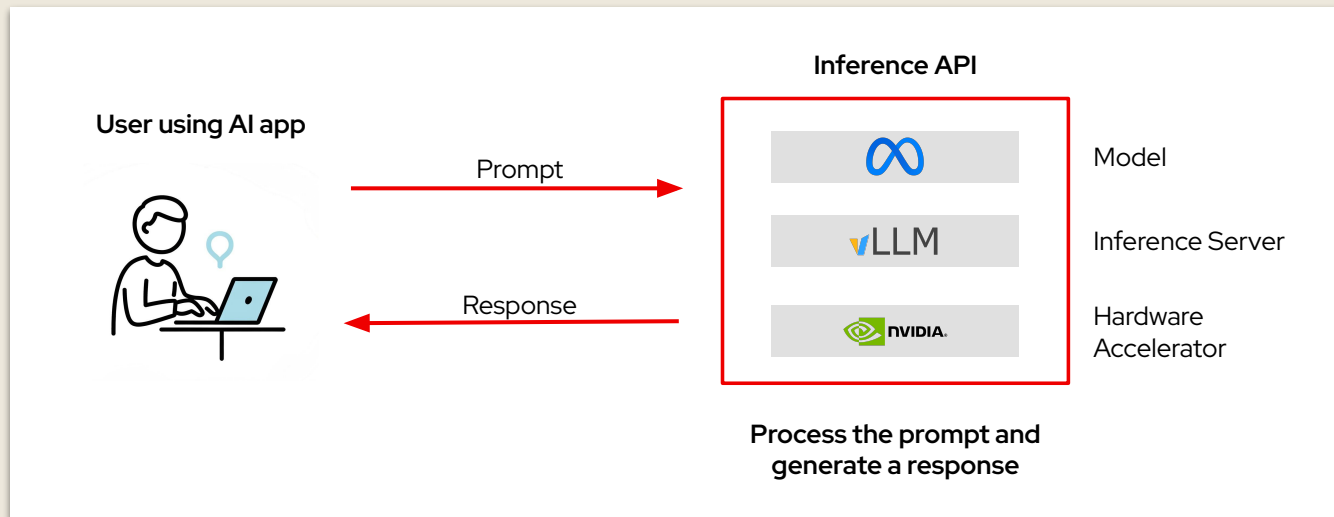
Demo video:

<https://youtube/j5HuRWWi5mA?si=uFMA7EPIMPLqJHg>

Demo - service mesh

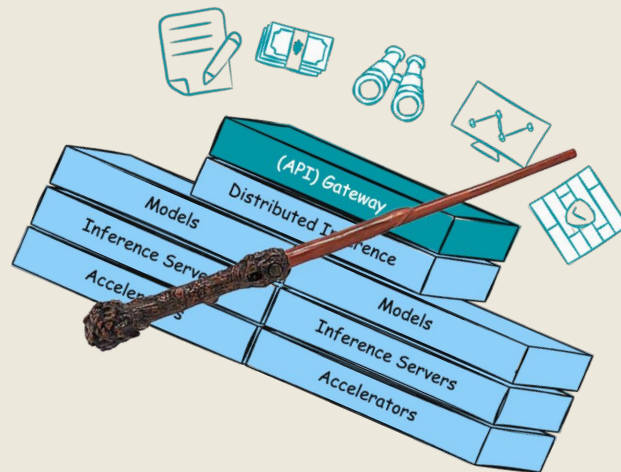
Demo - guardraíling

Demo - guarddrailing



Recap

- ✓ Think about sovereignty
- ✓ Monitor, monitor, monitor (maybe through monetization)
- ✓ Keep in control of your data and your models
- ✓ Reduce cognitive load
- ✓ Find out how to enter your Room of Requirement, how to start with MaaS
developers.redhat.com

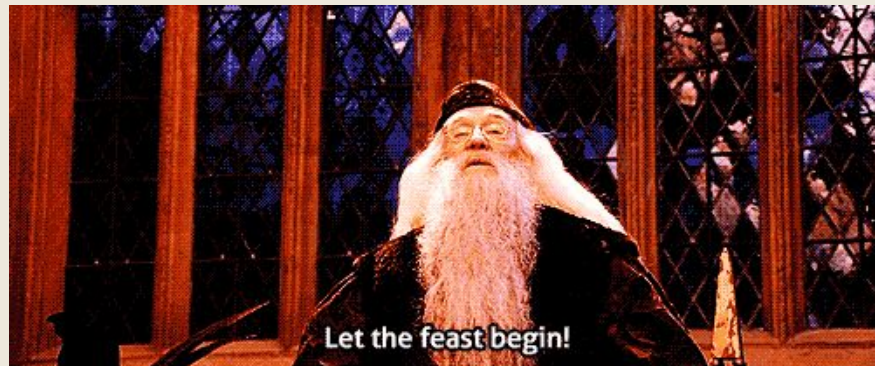


We turn developer work into wizardry. ✨

Digital sovereignty isn't that complex...

... within controlled boundaries!!!

... stay in control!!!



And...

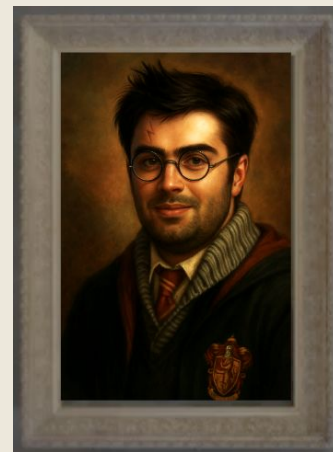
Keep your options open



Thanks!



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<https://maarten-vandepierre.be>



<https://github.com/camille-maarten/presentations>



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